

GOLDFISH · CARASSIUS AURATUS

The bowl was always wrong

A complete, fact-checked printable manual: tank size, filtration, cycling, water chemistry, safe foods, health triage, and the routines that keep a goldfish thriving for its full lifespan.

INSIDE

- ✔ Tank size by variety, with filtration turnover math
- ✔ Fishless and fish-in cycling, step by step
- ✔ Safe foods charts: pellets, greens, protein and the never list
- ✔ Owner logs, pet-sitter sheet and a power-outage plan

39 pages

Beginner

Print or tablet

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Every page is written to be used, not just read once.

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START HERE

How to use this package

Three ways to get value from it, depending on where you are right now.

If you haven't set up yet

Read pages 4 through 18 in order before buying anything. The tank and the filter are the two decisions that determine how the next decade goes, and both are far easier to get right the first time than to correct later with a fish already in the water. Budget for the tank before the fish, and expect to spend four to six weeks cycling it before a goldfish goes in. Then use the Budget and Shopping List on page 29 to buy once.

If your goldfish is already set up

Jump to the Setup Checklist and Targets on page 27 to audit what you have, and the Health Red Flags section starting on page 22 if anything seems off. Test the water first, before you change anything else. Most goldfish problems that look like disease are a water reading, and you cannot rule that out by looking at the tank.

If you already have a fish in an uncycled tank

This is the most common situation a goldfish owner is actually in, and it is fixable. Go straight to the fish-in cycle on page 9. Do not tear the tank down, do not replace the filter, and do not start dosing treatments. Daily testing and water changes carry you through it.

Print these first

- | | |
|--|---|
| ☐ Page 27: Setup Checklist & Targets | ☐ Page 34: Pet-Sitter Sheet |
| ☐ Page 28: Emergency & Quick Targets Card | ☐ Page 35: Owner Log |
| ☐ Page 31: Symptom Quick Reference | ☐ Page 36: Equipment & Maintenance Log |
| ☐ Page 32: Daily, Weekly & Seasonal Routine | ☐ Page 37: Enrichment Checklist & Log |

WHY THIS PACKAGE EXISTS

No pet is sold with worse instructions than a goldfish. The bowl on the counter, the "grows to fit its tank" line, and the three-second memory are all still repeated in stores, and every one of them is wrong. This package replaces them with numbers you can test against. Cross-checked against veterinary and aquarium-science sources (listed on page 39), not recycled forum advice. Every temperature and dimension is given in both imperial and metric.

NOT A SUBSTITUTE FOR

An aquatic veterinarian. This is a husbandry reference for day-to-day care. Anything urgent or health-related is always a professional call, not a page in this book. Aquatic vets are rarer than small-animal vets, so find one before you need one and write the number on page 28.

SECTION 01

Quick profile

Everything at a glance before you read the detail.

Scientific name	<i>Carassius auratus</i>
Adult size	Fancy varieties 6 to 8 in (15 to 20 cm); common and comet 10 to 14+ in (25 to 36+ cm). Pond fish have exceeded 18 in (46 cm)
Captive lifespan	10 to 15 years in a well-kept tank; 20 to 30+ years in a spacious pond. The record holder reached 43 years
Origin	Domesticated in China over a thousand years ago from a wild carp relative
Diet type	Omnivore. Sinking goldfish pellets as the staple, vegetables and protein foods alongside
Tank minimum (adult)	20 gal (76 L) for one fancy, plus 10 gal (38 L) per extra fish; 55 gal (208 L) for one common or comet
Water temperature	65 to 75°F (18 to 24°C). A coldwater fish; no heater needed at normal room temperature
Water chemistry	Ammonia 0, nitrite 0, nitrate under 20 ppm, pH 7.0 to 7.5. Full table on page 10
Filtration minimum	4 times the tank volume per hour, 5 to 10 times preferred
Sexual maturity	Roughly 9 to 12 months, which is also when sexing first becomes possible
Handling difficulty	Not a hands-on pet. Container transfers only, never bare hands (page 13)
Social needs	Social. Does well in same-type groups if volume and filtration support the bioload
Conservation status	Domesticated, not evaluated. The wild ancestor is Least Concern

TWO CLAIMS TO UNLEARN FIRST

A goldfish does not grow to fit its container, it becomes stunted in a small one, which shortens its life. And it does not have a three-second memory: goldfish form associative memories lasting months and can be trained. Both myths exist to make an inadequate setup sound acceptable. The full argument is on page 6.

SECTION 01

Cost & commitment

What the fish costs, and what the fish actually asks of you.

Initial setup

\$208 to \$500 for equipment, plus \$5 to \$30 for the fish itself. Full itemized list on page 29.

Monthly ongoing

\$13 to \$32 for food, water conditioner, filter media and electricity, averaged.

BUDGET FOR THE WORST CASE TOO

The goldfish is the cheapest animal in the store and one of the more expensive fish to house correctly. The real risk to your budget is not the monthly cost, it is replacing an undersized tank and filter six months in, once the water will not stay clean. Aquatic vets are uncommon and consultations are priced as a specialist service, so set aside a few hundred dollars, and buy the correct tank once rather than twice.

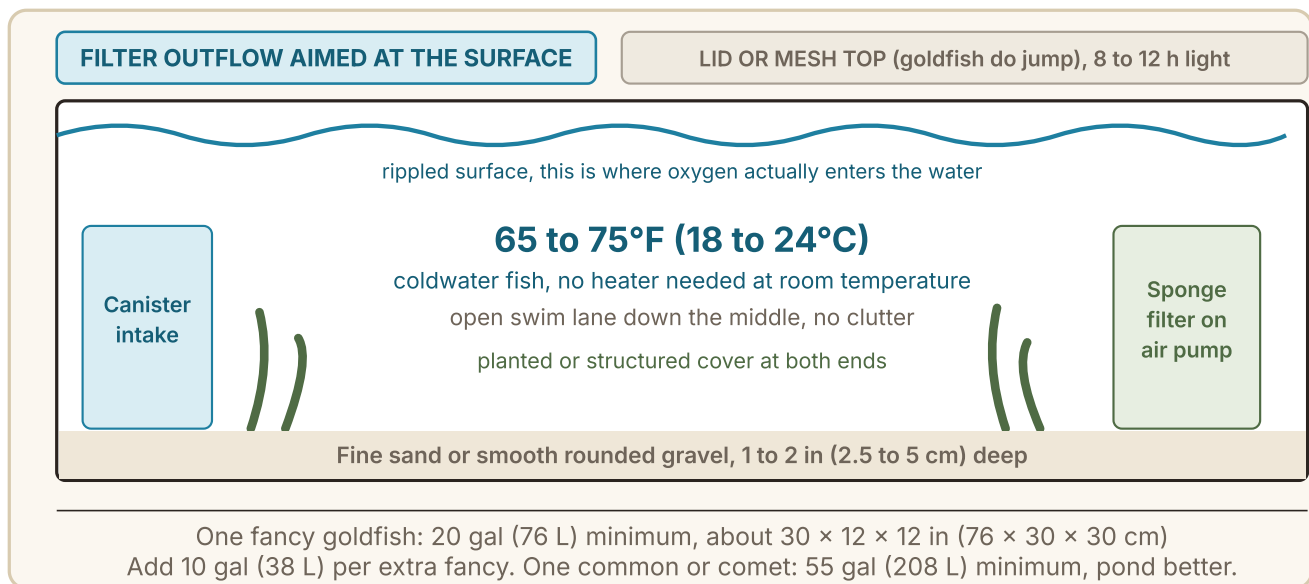
Fun facts worth knowing

- Goldfish form associative memories that last months. They can be trained to navigate mazes, push levers for food, and learn feeding routines, which makes the three-second memory line not just wrong but backwards.
- Goldfish have no true stomach. Food passes more or less straight through the intestine, which is exactly why small frequent portions work and one large meal does not.
- Goldfish and koi are different species in different genera and cannot interbreed, despite both being carp relatives selectively bred for color over centuries.
- A single goldfish produces a waste load comparable to several tropical fish of the same size combined, which is the whole reason the filtration numbers on page 7 look excessive for the animal.

Tank size & the bowl myth

A single fancy goldfish needs a minimum of 20 gal (76 L), with 10 gal (38 L) added for each fish after that. Single-tail varieties, the commons and comets sold as feeder fish, reach 10 to 14+ in (25 to 36+ cm) and need at least 55 gal (208 L) for one fish; long term they belong in an outdoor pond, not an indoor tank. Long and wide beats tall. Goldfish are horizontal swimmers, and floor area drives both swimming room and the surface area available for gas exchange.

Goldfish do not grow to fit their container. What actually happens in a small tank is stunting: chronic ammonia exposure and cramped space suppress growth hormone while the skeleton and organs keep developing, and the result is a shortened lifespan and organ failure, not a conveniently small fish. A bowl-kept goldfish often survives months or a couple of years, which is long enough for the setup to look like it works while the fish is slowly declining.



NEVER

Keep a goldfish in a bowl, a vase, or a self-cleaning aquaponics kit. A few plant roots cannot process the ammonia a goldfish produces, and none of those containers hold enough water or surface area for the fish to breathe properly. The marketing changed; the underlying problem did not.

What the diagram doesn't show

- A full 20 gal tank weighs roughly 225 lb (102 kg) once filled and gravelled. Check the stand and the floor before you fill it, not after.
- Cover the tank. Goldfish startle and can jump clear of an open rim, and a lid also slows evaporation so the water level stays stable between changes.

Filtration

A goldfish produces a bioload comparable to several tropical fish of the same size combined. That single fact drives almost every filtration decision here, and it is the reason experienced goldfish keepers run two filters as standard practice rather than as backup equipment.

FILTRATION TARGET	NUMBER
Minimum turnover	4 times the tank's total volume per hour
Preferred turnover	5 to 10 times per hour
On a 20 gal (76 L) tank	80 gph (300 lph) minimum, 100 to 200 gph (380 to 760 lph) better
On a 40 gal (151 L) tank	160 gph (605 lph) minimum, 200 to 400 gph (760 to 1,500 lph) better
On a 55 gal (208 L) tank	220 gph (830 lph) minimum, 275 to 550 gph (1,040 to 2,080 lph) better

Manufacturer gallons-per-hour ratings are measured with an empty filter and no head height. Assume the real figure is meaningfully lower once the baskets are full and the hoses are run, and size up accordingly. A filter rated for a tank twice your volume is a reasonable default for goldfish.

AIM THE OUTFLOW AT THE SURFACE

Oxygen enters the water at the surface, and a goldfish uses a lot of it. Angling the filter return so it visibly ripples or churns the surface does more for oxygenation than anything else in the tank, costs nothing, and matters most in summer, when warmer water carries less dissolved oxygen.

Filter media & maintenance

Three kinds of media, all three needed

MEDIA	WHAT IT DOES	HOW TO MAINTAIN IT
Mechanical sponge, floss, filter wool	Catches solid waste before it breaks down into ammonia.	Rinse monthly in removed tank water. This is the one you clean often, and the one that clogs.
Biological ceramic rings, bio-media, sponge	Houses the bacteria colony that converts ammonia to nitrite to nitrate. The actual engine of the tank.	Disturb as little as possible. Never replace it all at once. If it looks dirty, that is the colony.
Chemical activated carbon	Polishes water, removes tannins, strips out medication residue after treatment.	Optional in a normal tank. Replace monthly if used, and remove it while medicating.

NEVER

Rinse filter media under the tap. Chlorine and a sudden temperature change kill the beneficial bacteria colony that is doing the work, and a tank can crash within days of a well-intentioned deep clean. Rinse media in a bucket of water you just siphoned out of the tank, and replace only one filter's media at a time.

Why two filters, or a filter plus an air pump

Running a second filter, or a sponge filter on an air pump alongside the main unit, does three useful things at once for very little money. It adds biological capacity for a species that badly needs it. It means a failed impeller does not take out your entire bacteria colony overnight. And a battery-powered air pump on that sponge filter is the single most useful thing you can own during a power cut, covered on page 33. A mature sponge filter is also a ready-made instant cycle for a hospital or quarantine tank.

Cycling the tank

Cycling means growing a bacteria colony that converts ammonia to nitrite, then nitrite to nitrate. Until it exists, the tank cannot process waste and a fish put in it is living in its own toxins. This is the step that gets skipped, and skipping it has a name: new tank syndrome, the most common cause of premature goldfish death.

The fishless cycle, four to six weeks

STAGE	WHAT YOU'LL READ ON THE TEST KIT	WHAT TO DO
Week 1 to 2	Ammonia climbing, nitrite 0, nitrate 0	Dose ammonia to about 2 ppm, or add a pinch of fish food, and wait. Filter and air pump run continuously.
Week 2 to 4	Ammonia falling, nitrite climbing and often spiking hard	Keep dosing. The nitrite spike is the longest and most discouraging stage. Do not add fish.
Week 4 to 6	Ammonia 0, nitrite 0, nitrate rising	Cycled. Do a 50 percent water change to bring nitrate down, then add fish.

SPEEDING IT UP HONESTLY

A handful of used media, a squeeze of sponge, or a few cups of gravel from an established, disease-free tank seeds the colony and can cut weeks off the process. Bottled bacteria products vary widely in whether they help. Either way, the test kit decides when the tank is cycled, not the calendar and not the label.

The fish-in cycle

Plenty of goldfish are bought on the same trip as the tank, and if that is where you are, this is the recovery plan rather than a reason to start over. The goal is keeping ammonia and nitrite low enough not to harm the fish while the colony establishes, which takes roughly two to six weeks.

- **Test daily**, or at least every other day, for ammonia and nitrite.
- **Water change at 0.25 ppm.** Any reading above roughly 0.25 ppm of ammonia or nitrite calls for a 25 to 30 percent change with dechlorinated, temperature-matched water. Larger changes than that slow the colony you are trying to grow.
- **Feed lightly.** Small portions once a day, increasing gradually over the four to six weeks. Every uneaten pellet is more ammonia the tank cannot yet process.
- **Use an ammonia binder.** A conditioner that locks up free ammonia protects the gills without stalling the cycle, and is worth having on the shelf regardless.
- **Do not clean the filter**, replace media, or deep clean the substrate during a cycle. You would be removing the colony you are waiting for.

HOW YOU KNOW IT'S FINISHED

You can feed normal portions for a full week while ammonia and nitrite both hold at 0 ppm and nitrate reads above 0. Nitrate appearing is the proof the cycle is complete: it is the end product, and its presence means both earlier conversions are happening. From then on, nitrate is what your weekly water changes exist to remove, covered on page 11.

Water targets & testing

Every number on this page is testable. None of it can be judged by looking at the tank.

READING	TARGET	ACT WHEN
Ammonia	0 ppm	Anything above 0. Water change now.
Nitrite	0 ppm	Anything above 0. Water change now.
Nitrate	Under 20 ppm	Over 40 ppm. Larger or more frequent changes.
pH	7.0 to 7.5 ideal	Below 6.5 or above 8.0, or moving fast. Stability matters more than the exact figure.
Temperature	65 to 75°F (18 to 24°C)	Swings over about 4°F (2°C) in a day.
GH (general hardness)	100 to 300 ppm	Rarely a problem on typical tap water.
KH (carbonate hardness)	70 to 140 ppm	Under about 40 ppm. Low KH lets pH crash without warning.

Reading the numbers together

Ammonia and nitrite are the emergency pair. Both are acutely toxic and both should read exactly zero in a cycled tank. A reading above zero is not a trend to watch, it is an action today.

Nitrate is the housekeeping number. It is far less toxic and it accumulates continuously, so it measures how well your water change schedule is keeping up with your stocking and feeding. Climbing nitrate between changes means feeding more than the tank can absorb, too many fish, or changes that are too small.

pH and KH work as a pair. KH is the water's buffer, its resistance to pH movement. When KH falls too low, the acids produced by fish respiration and bacterial activity can drop the pH suddenly and dangerously. That is why a tank with soft, low-buffer water is worth watching even when the pH reading itself looks fine, and why chasing a pH number with adjusting chemicals so often makes things worse rather than better.

LIQUID KIT, NOT PAPER STRIPS

Strips are convenient and consistently less accurate, and the readings that matter most here are the two that have to read exactly zero. A liquid master kit costs more once and gives you numbers you can act on. Check the expiry date: reagents lose accuracy, and an out-of-date kit reading zero is not the same as a tank reading zero.

Water changes & early warnings

The weekly water change

Change 25 to 50 percent weekly on a stocked goldfish tank. Some general fish care sheets suggest smaller changes at longer intervals; for a goldfish specifically, the waste output justifies the weekly schedule, and nitrate creeping up between changes is the sign to do more rather than less.

1. Test first, and write the numbers down before you change anything.
2. Unplug any heater. Leave the filter alone; you are not cleaning it today.
3. Siphon from the substrate up, working across the floor of the tank. Most of what you are removing has settled into the sand or gravel, so the siphon is doing two jobs at once.
4. Fill a clean, tank-only bucket with tap water and dose dechlorinator for the whole bucket volume.
5. Check the new water is within a few degrees of the tank, then pour it back slowly, against the glass or onto a plate, so you do not blast the substrate.
6. Plug everything back in and confirm the filter has restarted and the surface is moving.

CHLORAMINE DOES NOT EVAPORATE

Many municipal supplies now use chloramine, a chlorine and ammonia compound, instead of plain chlorine. Unlike chlorine it does not gas off if you leave a bucket standing overnight, so the old advice to age water is not a substitute for a conditioner. Use a dechlorinator rated for chloramine, every time, including for small top-ups.

NEVER CHANGE EVERYTHING AT ONCE

A 100 percent change, a full substrate wash, or a scrub of every surface strips out the bacteria colony and restarts the cycle with a fish in the tank. If the water looks bad, the answer is several partial changes across a few days, not one drastic one.

Substrate, plants & decor

Substrate

Fine sand or smooth rounded gravel, 1 to 2 in (2.5 to 5 cm) deep. Research on substrate particle size found that coarse sand supported more natural foraging behavior than larger gravel, and goldfish kept sifting even when they were not hungry. A bare-bottom tank is easier to keep spotless and is a legitimate choice, but it removes a real foraging outlet that costs nothing to provide.

NEVER

Use sharp gravel, crushed glass decor, or any pebble small enough for a goldfish to fit in its mouth but too large to pass. Goldfish sift substrate constantly and a lodged stone is a genuine emergency. Either go finer than the fish can swallow, or larger than it can lift.

Plants

Expect any live planting to be grazed, uprooted, or both. Rhizome plants such as anubias and java fern, tied or glued to wood or rock rather than buried, hold up far better than anything rooted in the substrate. Tough, fast-growing stems and floating plants are the other workable approach, on the understanding that some get eaten. Grazing is not a problem to solve, it is normal feeding behavior, and soft plants like duckweed count as an occasional food rather than decor.

Artificial planting is a legitimate alternative here, and not a compromise. The habitat-preference research that found goldfish spending roughly 90 percent of their time in planted areas found no strong preference for live plants over closely matched artificial ones. It is the structure that matters. If you use artificial plants, choose silk or soft fabric rather than stiff plastic, which tears fancy fins.

Decor checklist

- | | |
|--|---|
| ☐ Smooth hardscape only, nothing that can catch a fancy fin | ☐ Planted or structured cover at both ends |
| ☐ Rhizome plants tied to wood or rock, not buried | ☐ Everything anchored so nothing can topple onto a fish |
| ☐ An open swim lane running the length of the tank | ☐ No stiff plastic plants, sharp ornaments, or holes a fish can wedge into |

TWO DECOR TRAPS SPECIFIC TO FANCY GOLDFISH

Ornaments with narrow openings, and anything with a hollow interior, catch a round-bodied fish that swims into them and cannot reverse out. And telescope-eye, bubble-eye and celestial varieties have fragile, protruding eyes with poor vision: for those fish, every surface in the tank needs to be smooth, and sharp hardscape is a real injury risk rather than a theoretical one.

Handling, quarantine & settling in

Handling: net and cup, never hands

A goldfish's slime coat is a living barrier against parasites, bacteria and infection, and bare hands strip it on contact. Losing it also disrupts the fish's ability to regulate salts and water balance in its own body, so it is a mechanical stress on top of the infection risk. The coat takes 24 to 48 hours to regenerate, and during that window the fish is meaningfully more vulnerable than usual. This is why a single unnecessary scoop by hand is not as harmless as it looks.

- Guide the fish into a container of its own tank water rather than lifting it out. This is the gentlest method and avoids the slime-coat problem almost entirely.
- If you must use a net, wet it first and choose soft rubber over knotted nylon, which catches fins and scales.
- If bare hands are genuinely unavoidable in an emergency, wet them first and stay clear of the gills and eyes. Minimize time out of the water in every case.

AQUARIUM WATER AND YOUR OWN HANDS

Aquarium water can carry *Mycobacterium marinum*, which causes a slow, stubborn skin infection through cuts and grazes. Wear waterproof gloves if you have broken skin on your hands, wash with soap afterward, and never start a siphon by mouth. Keep tank gear out of the kitchen sink.

Quarantine, and why it is worth the trouble

Almost every parasite and infection in this package arrives on a new fish, a new plant, or water from a shop tank. A quarantine tank is the one piece of equipment that prevents that, and it does not need to be elaborate: a bare 10 gal (38 L) tub, a sponge filter squeezed from your mature filter so it is instantly cycled, a heater if you need one for treatment, and a lid. Two to four weeks is the practical minimum, and four to six weeks is safer, since different pathogens surface on different timelines.

The same tank doubles as a hospital tank, which is where you want to be medicating rather than dosing your main tank: treatments are cheaper in a small volume, many of them harm your biological filter, and an ill fish is easier to observe without substrate and decor in the way.

First days in a new home

1. Float the sealed bag in the tank for 15 to 20 minutes so the temperature equalizes.
2. Open the bag and add a small amount of tank water every 5 minutes for another 20 to 30 minutes, so the fish adjusts gradually to your water chemistry.
3. Net the fish across and discard the bag water. Never pour shop water into your tank.
4. Lights off for the rest of the day, no feeding until the next morning, and no visitors at the glass.
5. Test ammonia and nitrite daily for the first week. A new fish is a new bioload and a small spike is common.

Feeding by age

Goldfish have no true stomach. Food passes more or less continuously through the intestine rather than being held and released, which is why small frequent portions suit them and a single large meal does not. It is also why overfeeding shows up so fast, as constipation, buoyancy trouble, and a waste load your filter was not sized for.

AGE	STAPLE	ALONGSIDE	FEEDINGS	HOW MUCH
Fry, first 2 days	Nothing	Nothing	None	Still absorbing the yolk sac. Feeding now only fouls the water.
Fry, free-swimming to 1 month	Infusoria, baby brine shrimp nauplii	Powdered fry food	3 to 6 a day	What is eaten in about a minute, siphon the rest.
1 to 4 months	Crushed sinking pellet, fry food	Baby brine shrimp, daphnia	About 3 a day	Small pinches, tapering as growth slows.
Juvenile, 4 to 12 months	Sinking goldfish pellet	Blanched vegetables 2 to 3 times a week	2 to 3 a day	Eaten in 30 seconds to 2 minutes per feeding.
Adult, 12 months on	Sinking goldfish pellet	Vegetables 2 to 3 times a week, protein treats weekly	1 to 2 a day	Eaten in 30 seconds to 2 minutes. Remove anything left.

Adult frequency is one of the few places good sources genuinely disagree, ranging from once a day to three times a day. One to two small meals is the safe middle, and splitting the daily ration into two feedings suits a fish with no stomach better than one large serving.

Portion, not schedule, is the thing to get right

Feed what disappears in 30 seconds to 2 minutes and siphon out the rest. A goldfish begs constantly and will keep eating well past the point of usefulness, so its behavior is not a portion guide, and neither is the feeding instruction on the tub. If the water is cloudy or nitrate is climbing between changes, you are feeding more than the tank can process.

THE FASTING DAY

Many experienced goldfish keepers skip feeding one day a week, particularly for round-bodied fancy varieties. A healthy adult goldfish is entirely comfortable going 3 to 4 days without food, so a weekly fast carries no risk and gives the gut a chance to clear. It is one of the simplest ways to reduce the constipation and buoyancy problems on page 25.

NO SUPPLEMENTS, AND THAT IS NORMAL

Unlike a reptile or an invertebrate, a goldfish needs no calcium dusting, no vitamin D3, and no gut-loaded feeders. Nutritional balance comes from genuine variety in the diet itself: a good pellet, vegetables two or three times a week, and protein foods in moderation. Anything sold as a goldfish supplement is doing a job the pellet should already be doing.

Pellets & presentation

Pellets, not flakes

A quality pellet formulated for goldfish is the base diet. Tropical flake is not toxic but its nutrient profile is built for a different fish, and flakes lose a meaningful share of some vitamins within about 30 seconds of hitting the water. Read the label for a named protein source near the front of the ingredient list rather than a generic filler, and buy a bag you will finish within a few months: opened fish food degrades, and a cheap large tub that lasts a year is a false economy.

Sinking, not floating, and why it matters more than it sounds

Sinking food matters especially for fancy varieties. Gulping floating pellets at the surface means swallowing air, and swallowed air is a direct contributor to the buoyancy problems round-bodied goldfish are prone to. Soaking pellets in a spoonful of tank water for a minute before feeding is a simple extra step that reduces both air intake and swelling in the gut, and it costs nothing. Gel foods sidestep the problem entirely: they sink, they carry a lot of moisture, and they suit fancy goldfish particularly well.

Scattering sinking food across the substrate rather than dropping it in one spot turns a 30 second meal into several minutes of foraging, and pairs with the sand or fine gravel on page 12. Same food, same portion, considerably more of the day spent doing something.

Feeding a pond goldfish by temperature

Outdoor fish scale their feeding to the water temperature, because their metabolism does. Feeding a normal ration to a fish whose digestion has slowed to a crawl leaves food sitting in the gut and in the water, and both cause problems.

WATER TEMPERATURE	WHAT TO FEED
Above 50°F (10°C)	Feed normally, scaled to how quickly it is taken.
39 to 50°F (4 to 10°C)	Switch to a reduced, more digestible wheat-germ formula, fed less often.
Below 39°F (4°C)	Stop feeding entirely. The fish drops into a low-metabolism dormancy and uneaten food simply fouls the water.

This applies to outdoor ponds only. An indoor tank at normal room temperature stays in the top row all year.

Staple foods & vegetables

The everyday base of the diet, and how to tell a good bag from a cheap one.

FOOD	TIER	NOTES
Sinking goldfish pellet	STAPLE	The base of the diet. Soak for a minute before feeding to cut air intake and swelling.
Gel food, goldfish formula	STAPLE	Excellent for fancy varieties. Soft, sinks, and carries a lot of moisture.
Goldfish flake	OCCASIONAL	Fine for small or young fish, but loses vitamins fast in water and floats. Not a long-term staple.
Tropical fish flake	AVOID AS STAPLE	Not toxic, but the nutrient profile is built for a different fish. Skip it as a primary food.
Wheat-germ pellet	OCCASIONAL	The cool-water pond formula. More digestible, and the right choice between 39 and 50°F (4 and 10°C).

FOOD	TIER	NOTES
Shelled peas, cooked and de-shelled	STAPLE	The single most useful vegetable here. Also the standard short-term aid for constipation.
Zucchini, blanched	STAPLE	Weight a slice down so it stays on the bottom. Remove after a few hours.
Cucumber, blanched or raw slice	OCCASIONAL	Low nutritional density but well accepted. Good foraging item.
Spinach, blanched	OCCASIONAL	Nutritious, but high in oxalates, so rotate rather than feed daily.
Kale, blanched	OCCASIONAL	Tough leaf, blanch until it softens fully or it will be shredded and left.
Lettuce, romaine or leaf	OCCASIONAL	Mostly water. Fine as forage, not as nutrition. Skip iceberg entirely.
Broccoli, blanched floret	OCCASIONAL	Well accepted. Chop small; the stalk is usually ignored.
Green beans, blanched and chopped	OCCASIONAL	Good fiber. Chop to a size the fish can actually manage.
Squash or pumpkin, blanched	OCCASIONAL	Softens well and readily taken. Useful alongside peas for a constipated fish.
Sweet potato, boiled	RARE	Starchy. Small pieces only, and not as a regular vegetable.
Algae wafers	OCCASIONAL	Sinks and lasts, which suits foraging. Once or twice a week.
Duckweed and other soft aquatic plants	OCCASIONAL	Goldfish graze these readily. Expect any live planting to be eaten.
Blanched nori or seaweed sheet, unseasoned	OCCASIONAL	Clip a strip to the glass. Plain only, never the salted or oiled snack kind.

Protein foods

Protein foods are variety and enrichment, not a second staple. Weekly at most for an adult.

FOOD	TIER	NOTES
Bloodworms, frozen	OCCASIONAL	Rich and eagerly taken. Once a week at most; thaw in tank water before feeding.
Brine shrimp, frozen	OCCASIONAL	Good general variety food, and gentler than bloodworms.
Brine shrimp, live	OCCASIONAL	Doubles as foraging enrichment, since the fish has to hunt it.
Daphnia, frozen or live	OCCASIONAL	High fiber for a protein food, so it is gentler on the gut. Useful for a constipated fish.
Mysis shrimp, frozen	OCCASIONAL	Well accepted and easy to portion. Thaw first.
Baby brine shrimp nauplii	FRY STAPLE	The standard first food for free-swimming fry. Not needed for adults.
Earthworm, chopped, from a clean source	OCCASIONAL	Genuinely good food if cultured or from untreated soil you know. Chop to size.
Blackworms or tubifex, live	TREAT ONLY	Very rich, and a real disease-introduction risk from unreliable sources.
Freeze-dried bloodworms	TREAT ONLY	Expands in the gut. If you use them, soak in tank water first, and rarely.
Freeze-dried brine shrimp	TREAT ONLY	Same expansion problem. Soak thoroughly or skip in favor of frozen.
Krill or plankton, frozen	RARE	Very rich. Small amounts, and mainly useful for conditioning a pond fish.
Live feeder fish	NEVER	A common route for introducing parasites and disease, and nutritionally pointless here.
Wild-caught insects, worms or crustaceans	NEVER	Pesticide and parasite risk you cannot assess. Buy cultured food instead.
Mammal or bird meat, in any form	NEVER	Those fats are not digestible for a goldfish and foul the water fast.

THE RICH-FOOD TRAP

Bloodworms and blackworms are the two foods goldfish respond to most dramatically, which is exactly why they get overfed. A fish that ignores pellets and comes alive for bloodworms is not telling you it needs more protein, it is behaving like an animal offered dessert. Keep protein foods to roughly once a week for an adult, and never use them to tempt a fish that has gone off its food: appetite loss is a symptom to investigate, on page 22, not a preference to indulge.

Treats, extras & the never-feed list

Fruit and human food are where most well-meant feeding goes wrong.

ITEM	VERDICT	NOTES
Orange, small piece	RARE	Occasionally offered and generally tolerated, but acidic. Not a routine food.
Watermelon, small piece	RARE	Mostly water and sugar. Novelty only, and remove it quickly.
Grapes or berries	RARE	Peel and quarter if you offer them at all. Sugar-heavy and easy to overdo.
Banana, small piece	RARE	Breaks apart and clouds the water quickly. More mess than benefit.
Apple, peeled and thin	RARE	Tolerated in tiny amounts. Never with seeds or core.
Boiled rice or plain pasta	RARE	Not toxic, but nutritionally empty and it clouds the water. Little reason to bother.
Bread, crackers, cereal	NEVER	Swells in the gut and can cause a blockage. The classic pond-feeding mistake.
Avocado	NEVER	Contains persin, which is potentially deadly to fish.
Chocolate	NEVER	Toxic in tiny amounts.
Onion, garlic, chives	NEVER	Irritant, and not part of any fish's diet.
Cheese, milk, yogurt	NEVER	Fish cannot digest dairy, and it fouls water immediately.
Raw or cooked meat and poultry	NEVER	Mammal and bird fats are indigestible for a goldfish.
Citrus as a habit	NEVER	Repeated exposure irritates gills and can disrupt tank pH.
Anything salted, seasoned, oiled or fried	NEVER	No safe version of this exists for an aquarium.
Dog or cat food	NEVER	Formulated for a mammal. Wrong protein, wrong fat, and it disintegrates in water.

THE NEVER-FEED LIST, IN ONE PLACE

Bread and other baked goods, avocado, chocolate, onion and garlic, dairy, meat and poultry, salted or seasoned human food, habitual citrus, dog and cat food, live feeder fish, and wild-caught invertebrates. Nothing on this list becomes safe in a small amount or as an occasional treat.

FEEDING A GOLDFISH IN A PUBLIC POND

The bread thrown at park ponds is the single most common way people harm goldfish and koi they do not own. If you want to feed pond fish, take a handful of the sinking pellets from page 16 instead. It costs the same, it does not swell in the gut, and it does not leave a mat of decaying starch behind in water nobody is filtering.

Common mistakes & enrichment

Mistakes that cause the most harm

- **Adding fish to an uncycled tank.** New tank syndrome is the single most common cause of premature goldfish death. There is no shortcut that beats testing until ammonia and nitrite both read zero.
- **Under-filtering because the tank looks clean.** Clear water and safe water are unrelated. Ammonia and nitrite are invisible, and a goldfish generates far more of them than its size suggests.
- **Overfeeding.** Begging is not hunger in this species, and every uneaten pellet becomes ammonia. Portion by the clock, not by the fish's enthusiasm.
- **Buying the fish before the tank.** The goldfish costs a few dollars and the setup costs a few hundred. Doing it in that order is what produces the bowl.
- **Cleaning too thoroughly.** Scrubbing every surface and rinsing media under the tap strips out the bacteria colony and can crash a stable tank in days.
- **Mixing goldfish with tropical fish.** A goldfish needs cool water and a tropical community needs a heater. One of the two is always being kept wrong.

Enrichment, by priority

Enrichment comes after correct volume, filtration, and a fully cycled tank, never instead of them. Decor cannot rescue a bowl or bad water. In this order:

1. Swimming volume

The single largest welfare factor. A long, wide footprint with a genuinely open lane down the middle beats any amount of decor in a small tank.

2. Structure and cover

Research tracking where goldfish spend their time found roughly 90 percent in planted compartments versus about 10 percent in barren open space, with no strong preference for live plants over closely matched artificial ones. It is the structure that matters.

3. Foraging substrate

Coarse sand supported more natural foraging than larger gravel in substrate research, and goldfish kept sifting even when food was not the only driver.

4. Sinking, scattered food

Scattering sinking pellets across the substrate turns one meal into several minutes of foraging. Same food, same portion, much more of the day spent doing something.

5. Same-type company

Goldfish are social. Match fancy with fancy and single-tail with single-tail so body shape and swim speed stay compatible. Only add fish if volume and filtration genuinely support the combined bioload.

6. Training

Goldfish learn readily and are strongly food-motivated. Target feeding, swimming through a hoop, and following a finger are all achievable, and use the daily ration rather than adding food.

NOT ENRICHMENT

Overstocking labeled as company, plants used as an excuse for weak filtration, and extra food framed as a treat are the same mistake wearing three different names. None of them substitute for getting the tank right.

Varieties, tankmates & sexing

Which goldfish do you actually have

TYPE	ADULT SIZE	MINIMUM FOR ONE	NOTES
Fancy: oranda, ranchu, fantail, ryukin, moor	6 to 8 in (15 to 20 cm)	20 gal (76 L), plus 10 gal (38 L) each	Round-bodied and comparatively poor swimmers. Most prone to buoyancy trouble.
Telescope-eye, bubble-eye, celestial	5 to 8 in (13 to 20 cm)	20 gal (76 L), plus 10 gal (38 L) each	Fragile protruding eyes and poor vision. Smooth surfaces only, and never with faster fish.
Shubunkin	10 to 12 in (25 to 30 cm)	40 gal (151 L)	Single-tail body on ornamental coloring. Faster and larger than it looks when sold.
Common and comet	10 to 14+ in (25 to 36+ cm)	55 gal (208 L), pond better	The fish sold as feeders. Fastest growing and least suited to an indoor tank long term.

DO NOT MIX BODY TYPES

A fancy goldfish and a comet in the same tank is not a community, it is a race the fancy loses at every meal. Single-tails are faster, larger and more competitive, and they will strip the food before a round-bodied fish reaches it. Match fancy with fancy and single-tail with single-tail. The eye varieties above need matching more carefully still, since they cannot see the food coming.

Telling male from female

Sexing a goldfish is genuinely unreliable outside of breeding condition, and even experienced keepers get it wrong. Signs only appear once the fish is sexually mature, at roughly 9 to 12 months, and are clearest in spring. Read several signs together rather than trusting any one of them.

- **Breeding tubercles.** Small white dots like grains of salt on the gill plates and the leading rays of the pectoral fins, rough to the touch like fine sandpaper. Usually male, but some males never develop them and, rarely, a female does.
- **Vent shape.** In breeding condition, females tend toward a rounder, slightly protruding vent; males toward a smaller, flatter, more slit-like one.
- **Body shape.** A gravid female often looks fuller and asymmetrical viewed from above. Males tend to be slimmer, and they are the ones doing the chasing during spawning.

THE ONLY CERTAIN METHOD

Observing an actual spawning. Everything else is a probability, and tubercles can be mistaken for the early white spots of ich, covered on page 23. Tubercles are raised, rough and confined to the gill plates and pectoral fin rays; ich spots are scattered across the body and fins and come with flashing and clamped fins. If you are buying with breeding in mind, buy a group of five or six rather than trying to pick a pair.

Growth, body condition & lifespan

Growth reference

Ranges are deliberately wide and healthy fish fall outside them. Growth is fastest in the first year, then slows to roughly an inch (2.5 cm) a year until full size at one to three years. Space, water quality and diet all cap it.

AGE	BODY LENGTH, NOSE TO TAIL BASE
Free-swimming fry to 2 months	0.2 to 1 in (0.5 to 2.5 cm)
2 to 12 months	1 to 4 in (2.5 to 10 cm), the fastest growth of its life
1 to 3 years	4 to 8 in (10 to 20 cm) for a fancy; single-tails run larger
Full size, fancy varieties	6 to 8 in (15 to 20 cm)
Full size, common and comet	10 to 14+ in (25 to 36+ cm)

Fish are measured nose to the base of the tail, not to the tail tip. A comet's trailing fins add several inches to what you see and none of it to the standard measurement.

Why we don't ask you to weigh a goldfish

For most pets a weekly weight is the best early warning available. For a fish it is the wrong tool: weighing means netting the fish out, which damages the slime coat for the sake of a number that moves with how recently it ate. Your equivalent vital signs are the water readings on page 10, taken weekly and logged on page 35, plus a photograph every few months to compare body shape over time. Both give you a trend, and neither touches the fish.

Body condition at a glance

UNDERWEIGHT

Visible spine ridge, pinched or hollow belly, head looking large for the body. Check for parasites and for competition at feeding time.

IDEAL

Smooth convex curve from head to tail, symmetrical from above, fins carried open and away from the body.

OVERWEIGHT OR BLOATED

Belly bulging past the body line, sluggish, buoyancy off. Distinguish fat from the sudden swelling of dropsy on page 24.

What actually determines lifespan

Ten to fifteen years is normal for a tank goldfish, twenty to thirty is routine in a spacious pond, and the record holder reached 43. The gap between a three-year goldfish and a fifteen-year one is almost entirely husbandry rather than luck: tank volume, filtration capacity, and a consistent water change schedule account for most of it, with diet and stocking density behind them. Cheap feeder-grade fish are bred for volume rather than health and start with a disadvantage, but even they routinely outlive their reputation in a correctly sized, well-filtered tank.

Health red flags

What to watch for, and what to actually tell the vet.

TEST THE WATER FIRST, THEN ACT, IF YOU SEE

Gasping at the surface or laboring gills. Red or purple streaking in the fins or bloodshot gills. Scales standing out from the body in a pinecone pattern. Floating, sinking, or swimming at a persistent angle. Cotton-like white growths. Open sores or red ulcers. White spots like grains of salt. Clamped fins with lethargy. Refusing food for 24 to 48 hours alongside any other sign on this list.

What to tell the vet

- ☐ Your last ammonia, nitrite, nitrate and pH readings, with dates
- ☐ Tank volume, filtration, stocking, and how long it has been running
- ☐ Water change size and frequency, and which conditioner you use
- ☐ When symptoms started and whether they're worsening
- ☐ Any new fish, plants or decor added in the last month
- ☐ Everything you have already dosed, with amounts and dates, and a photo or short video of the behavior

Why water quality comes up first, every time

Most goldfish disease is opportunistic. Ich, fin rot, fungus and ulcers all take hold in fish already stressed by ammonia, nitrite, crowding or a temperature swing, which is why treating the symptom without fixing the water usually buys a few weeks at most. Test before you medicate. Dosing a tank with the wrong treatment adds a stressor to a fish that already has one, and some medications also kill the filter's bacteria colony, which turns a disease problem into a water problem as well. Aquatic veterinarians are less common than small-animal vets, so find one before you need one and write the number on page 28.

Normal vs. concerning

NORMAL	CONCERNING
Resting on the bottom at night, or after lights out	Sitting on the bottom during the day with fins clamped
Occasional surface gulping right after feeding	Sustained gasping at the surface, gills working hard
Chasing and nudging in spring, in a mixed group	One fish being pursued relentlessly, with torn fins or scale loss
Trailing a normal-colored stool that drops away	Long, white, stringy trailing waste over several days
Color slowly deepening or fading with age and light	Sudden dark patches, especially after an ammonia spike

Ammonia poisoning & ich

Ammonia and nitrite poisoning

Cause: An uncycled tank, a tank overstocked for its filtration, overfeeding, or a filter crash after media was rinsed in tap water. It is the most common cause of premature goldfish death and it is entirely preventable. **Signs:** Gasping at the surface, red or purple streaks in the fins, bloodshot or pale gills, lethargy, clamped fins, refusing food, and dark patches on the body as ammonia burns heal. **Diagnosis:** A test kit, not the fish's appearance. Any reading above 0 ppm for ammonia or nitrite is the diagnosis. **Response:** A 50 percent water change immediately with dechlorinated, temperature-matched water, then repeat daily until both read 0. Stop feeding for 24 to 48 hours to cut the ammonia source. Increase surface agitation. Do not deep clean the filter, which would remove what little bacteria colony you have. **Recovery:** Gill damage and fin streaking take days to weeks to resolve once the water is right. Severe exposure can leave permanent gill damage.

THE ONE PRODUCT WORTH HAVING ON THE SHELF

An ammonia-binding water conditioner buys you hours during a spike by converting free ammonia to a form the fish cannot absorb while the bacteria colony catches up. It is a stopgap for an emergency, not a substitute for a water change or for enough filtration.

Ich, or white spot disease

Cause: The protozoan parasite *Ichthyophthirius multifiliis*, almost always introduced on a new fish, plant, or unquarantined decor, and taking hold in fish stressed by water quality. **Signs:** Tiny white spots like grains of salt across the body and fins, flashing (rubbing against surfaces), clamped fins, and rapid gill movement if the gills are affected. **Diagnosis:** Visually distinctive, though worth distinguishing from the finer, dustier sheen of velvet and from the raised white tubercles on a male in spring, covered on page 20. **Response:** Correct the water first, then treat the whole tank, never the individual fish, since the parasite is in the water and substrate as well.

WHY TREATMENT HAS TO OUTLAST THE SPOTS

Only the free-swimming stage of ich is vulnerable to treatment. While it is embedded in the fish or encysted on the substrate it is chemically untouchable, so treatment has to run long enough to catch every parasite as it becomes free-swimming. That cycle is measured in days for warmwater fish but in weeks for a coldwater fish like a goldfish. Merck's veterinary guidance is constant chemical exposure for at least 3 weeks; continue for at least 2 weeks after the last visible spot, whichever is longer.

Raising the temperature shortens the parasite's cycle, but goldfish are a coldwater species and warm water carries less oxygen, so this is a smaller lever here than in a tropical tank. If you use it, raise gradually over 24 to 48 hours to no more than about 78°F (26°C), add an air stone, and watch for gasping. Aquarium salt is a common adjunct and goldfish tolerate it well; follow the product's dosing or your vet's, add it dissolved and in stages, and remember it leaves the tank only through water changes.

Fin rot, fungus & dropsy

Three conditions that look unrelated and share one underlying cause.

Fin rot

Frayed, ragged or receding fin edges, often with a white or reddened margin, and usually bacterial. It follows poor water quality, an injury, or fin nipping in a badly matched group. Correcting the water is as much of the treatment as any medication: test, do a substantial water change, and keep up a schedule of them. If the erosion is advancing into the body of the fin after several days of clean water, an antibacterial treatment is warranted, and a vet's input is worth having before you dose an established tank. Fins regrow, though the new tissue often comes back paler or slightly misshapen.

Fungal infections

Cotton-like white or grey tufts, most often on a spot that was already injured or on a fish that has been stressed. Fungus is a secondary problem almost by definition: it takes an opening to get established. That makes prevention mostly a matter of avoiding unnecessary handling, keeping decor smooth, and not letting a minor injury sit in dirty water. Antifungal treatments exist and work; the more important step is finding the injury or stressor that let it in.

Dropsy

A swollen belly with scales standing out from the body in a pinecone pattern, most visible looking down from above. Dropsy is not a disease in itself, it is a visible sign of fluid accumulation, usually from organ or kidney failure, and it frequently carries a poor prognosis. It is not a condition to experiment on with over-the-counter products. Get expert or veterinary help promptly, isolate the fish if you have a cycled hospital tank ready, and keep the water pristine in the meantime.

PINECONE SCALES ARE AN EMERGENCY, NOT A WAIT-AND-SEE

By the time scales are visibly raised, the underlying problem has usually been developing for a while. Fish caught at the very earliest stage sometimes recover with veterinary treatment. Fish left another week generally do not.

THE COMMON THREAD

All three of these are opportunistic. The bacteria and fungi involved are present in every aquarium at all times and cause nothing in a healthy fish in clean, stable, well-oxygenated water. They become infections when something else has already weakened the fish: an ammonia spike, a temperature swing, an injury, crowding, or a badly matched tankmate. This is why the first response to every condition in this section is a test kit rather than a bottle, and why the routine on page 32 does more for your fish's health than anything on a pharmacy shelf.

Swim bladder disorder & minor conditions

Less dramatic than the last page, and far more common.

Swim bladder disorder and constipation

Buoyancy trouble: floating at the surface, sinking to the bottom, or swimming at a persistent angle or upside down. It is much more common in round-bodied fancy varieties, whose compressed body shape crowds the swim bladder and the gut. The usual triggers are diet-related: overfeeding, constipation, and gulping air at the surface while taking floating food. Start with 24 to 48 hours of fasting, then a single cooked, de-shelled pea, and switch permanently to sinking or pre-soaked food. Not every case resolves with diet, and a fish that stays affected for more than a week or two, or that also has swelling or reddening, needs a vet: infection, an internal tumor and physical injury all produce the same outward sign.

Flukes, gill and skin

Microscopic monogenean worms causing flashing, rapid gill movement, excess mucus and irritation with nothing visible on the fish. They are easy to miss and frequently the answer when a fish behaves like it has ich but has no spots. Merck lists praziquantel at 5 mg/L as a prolonged bath as the treatment of choice for ornamental fish, repeated at weekly intervals because the eggs resist treatment.

Anchor worm and fish lice

Visible parasites. Anchor worm (*Lernaea*) appears as a thread-like projection with a red, inflamed attachment point; fish lice as flat, disc-shaped crawlers. Both usually arrive with pond fish or unquarantined stock. Merck gives diflubenzuron at 0.03 mg/L once as most effective, with a 3 percent (30 g/L) salt dip of under 10 minutes followed by salt at 5 g/L in the tank for 3 weeks as an alternative for freshwater fish. Do not pull an anchor worm out without veterinary guidance; the embedded head tears the tissue and leaves a wound.

Velvet

A fine gold or rust-colored dust over the skin, best seen with a flashlight held at an angle in a darkened room. Finer and more uniform than ich's distinct spots. Treat the whole tank and dim the lights, since the organism is partly photosynthetic.

Popeye and ulcers

A protruding eye is either an injury, usually in one eye, or a systemic infection, often in both. Open red sores or raised reddened patches are bacterial ulcers, and they follow the same water quality story as fin rot. Both warrant a vet rather than a shelf product, particularly when more than one fish is affected.

MEDICATE IN A HOSPITAL TANK, NOT THE DISPLAY TANK

Most of the treatments named on this page are cheaper in a small volume, easier to dose accurately, and safer for your biological filter, several of them kill filter bacteria outright. The exception is ich, which lives in the substrate and water as well as on the fish, so the whole tank has to be treated. Keep a cycled sponge filter running spare, as on page 13, and a hospital tank is twenty minutes of setup rather than a project.

Behavior, spawning & reading waste

Spawning behavior in spring

As temperatures rise in late spring, mature goldfish in a mixed group may spawn. Males chase a female persistently, often pinning her against the glass or plants, and both may go off food for a few days. This is hormone-driven and normal rather than aggression, and it usually resolves in days to a couple of weeks. Watch for a female being pursued to exhaustion or losing scales, which is the point to separate the group. Eggs are scattered over plants and surfaces and will mostly be eaten by the adults, which is the normal outcome unless you deliberately set up to raise fry.

Winter slowdown

Pond goldfish drop into a low-metabolism dormancy as water cools, settling near the bottom and eating little or nothing below about 39°F (4°C). This is normal, and it does not apply to an indoor tank held at room temperature. An indoor goldfish that suddenly becomes sluggish has a water quality or health problem, not a season.

Behaviors people misread

BEHAVIOR	WHAT IT USUALLY MEANS
Flashing, rubbing against surfaces	Irritation. Test the water, then look for ich, velvet or flukes.
Sifting substrate and spitting it out	Normal foraging. This is what the substrate is for.
Hanging at the surface gulping	Low oxygen or gill damage. Test immediately and add surface agitation.
Sitting on the bottom at night	Sleeping. Goldfish rest with reduced color and stillness, and have no eyelids.
Chasing at the surface at feeding time	Learned anticipation. Goldfish associate people and routines with food within days.
Yawning, or a wide slow mouth gape	Normal. Fish stretch their jaws and clear their gills this way.

Stool is the earliest free health check you get. Look at it before you siphon it up.

IT LOOKS LIKE	USUALLY MEANS	DO
Short, dark, matching the food, dropping away cleanly	Normal	Nothing. Siphon it out at the next water change.
Long, pale, stringy and trailing for days	Constipation, overfeeding, or too much dry food	Fast 24 to 48 hours, then a de-shelled pea. Switch to soaked or sinking food.
White, stringy and clear rather than pale	Often internal parasites, especially alongside weight loss with a good appetite	Vet or a targeted antiparasitic. Do not treat blind.
Bubbly, gassy, or with visible air	Air swallowed at the surface with floating food	Move to sinking or pre-soaked pellets permanently.
Bloody or red-tinged	Internal inflammation or injury	Test water, then vet. Not a wait-and-see sign.
Bright green or red, matching a recent food	Usually just the pigment of peas or bloodworms passing through	Nothing, if it is short-lived and the fish is otherwise well.

Setup checklist & targets

Essential setup checklist

- ☐ 20 gal (76 L) minimum per fancy goldfish, plus 10 gal (38 L) each additional. 55 gal (208 L) for a common or comet
- ☐ Filtration rated for 4 times the tank volume per hour minimum, 5 to 10 times preferred
- ☐ Air pump and sponge filter as backup biofiltration and outage insurance
- ☐ Liquid test kit: ammonia, nitrite, nitrate, pH
- ☐ Dechlorinator rated for chloramine
- ☐ Fine sand or smooth rounded gravel, 1 to 2 in (2.5 to 5 cm) deep
- ☐ Gravel vacuum and two buckets used only for the tank
- ☐ Soft rubber net and a transfer container
- ☐ Thermometer, and a lid or mesh top
- ☐ Sinking goldfish pellets. No heater needed at room temperature

Water targets

READING	TARGET	ACT WHEN
Ammonia	0 ppm	Above 0
Nitrite	0 ppm	Above 0
Nitrate	Under 20 ppm	Over 40 ppm
pH	7.0 to 7.5	Below 6.5 or above 8.0, or moving fast
Temperature	65 to 75°F (18 to 24°C)	Swings over about 4°F (2°C) in a day
KH	70 to 140 ppm	Under about 40 ppm
Water change	25 to 50% weekly	Nitrate climbing between changes
Light cycle	8 to 12 hours a day	Algae taking over, or no dark period at all

Before your goldfish comes home

- ☐ Tank fully cycled: ammonia 0, nitrite 0, nitrate present. Four to six weeks, confirmed by test kit
- ☐ Filter and air pump running continuously for at least a week
- ☐ Temperature stable and within a few degrees of the store's water
- ☐ Aquatic vet located and the number written on page 28

Emergency & quick targets

TEST THE WATER, THEN CALL AN AQUATIC VET, IF YOU SEE

Scales standing out in a pinecone pattern. Gasping at the surface with laboring gills. Red or purple streaking in the fins or bloodshot gills. Open sores or red ulcers. Persistent floating, sinking or swimming upside down for more than a few days. A protruding eye. Refusing food for 24 to 48 hours alongside any other sign here. More than one fish sick at once.

READING	TARGET	EMERGENCY ACTION
Ammonia	0 ppm	50% water change now, repeat daily
Nitrite	0 ppm	50% water change now, repeat daily
Nitrate	Under 20 ppm	Larger or more frequent changes
pH	7.0 to 7.5	Never correct fast. A stable wrong number beats a moving right one
Temperature	65 to 75°F (18 to 24°C)	Match new water to within a few degrees

Your numbers

Aquatic vet _____

Tank volume and date set up _____

24-hour emergency clinic _____

Filter model and media changed _____

Local fish store _____

Test kit expiry date _____

HAVE THIS READY WHEN YOU CALL

Your four most recent test readings with dates, the tank's volume and filtration, how many fish are in it, everything you have already added to the water, and a short video of the behavior. A photo of a fish in a bag or a bare white container shows color and scale detail far better than one shot through aquarium glass.

EMERGENCY WATER CHANGE, THE SHORT VERSION

Fill a clean bucket, add dechlorinator at the dose for the whole bucket, check the temperature is within a few degrees of the tank, siphon out half the tank from the substrate up, then refill slowly. Do not remove the fish. Do not clean the filter on the same day.

Budget & shopping list

US retail ranges. The low end is basic but correct gear; the high end is premium equipment.

INITIAL SETUP	TYPICAL COST
20 gal (76 L) long tank, the fancy goldfish minimum	\$40 to \$120
Filtration: canister or two hang-on-back units, 4x turnover minimum	\$45 to \$130
Air pump, airline and sponge filter (backup biofiltration and outage insurance)	\$20 to \$40
Fine sand or smooth rounded gravel substrate	\$15 to \$30
Liquid test kit: ammonia, nitrite, nitrate, pH	\$25 to \$40
Dechlorinator	\$8 to \$15
Gravel vacuum and two dedicated buckets	\$15 to \$30
Plants, hardscape and smooth decor	\$15 to \$45
Soft rubber net and a transfer container	\$10 to \$20
Thermometer and a lid or mesh top	\$5 to \$10
First bag of sinking goldfish pellets	\$10 to \$20
Equipment subtotal	\$208 to \$500
The fish itself, more for fancy varieties	\$5 to \$30
All-in before the goldfish arrives	\$213 to \$530

MONTHLY ONGOING	TYPICAL COST
Sinking pellets, frozen or gel food, and vegetables	\$6 to \$15
Dechlorinator and test kit refills	\$3 to \$7
Filter media replacement, averaged over the year	\$2 to \$6
Electricity for filter and air pump, averaged	\$2 to \$4
Monthly ongoing	\$13 to \$32

PLAN FOR, DON'T COUNT ON AVOIDING

A replacement filter or impeller, a hospital or quarantine tank for a sick or new fish, medications, and an aquatic vet consultation, which is a specialist service and priced like one. A few hundred dollars set aside matters as much as any item on this list.

WHERE NOT TO CUT CORNERS

The tank and the filter, in that order. Everything else here can be upgraded later without disturbing the fish; those two cannot. The liquid test kit is third. Cutting tank size is the one saving that reliably costs more, because the replacement tank still gets bought, just later and alongside a sick fish.

First 30 days checklist

Before day 1

- ☐ Tank cycled and confirmed: ammonia 0, nitrite 0, nitrate present
- ☐ 50 percent water change done to bring nitrate down before the fish arrives

Day 1

- ☐ Float the sealed bag 15 to 20 minutes, then add tank water to it every 5 minutes for 20 to 30 minutes before netting the fish across
- ☐ Photograph the fish from the side and from above as a baseline for color, fins and body shape
- ☐ Lights off for the rest of the day, no feeding until tomorrow morning
- ☐ Fill in the numbers on page 28 and the equipment dates on page 36

Week 1

- ☐ Test ammonia and nitrite daily. A new fish is a new bioload and a small spike is common
- ☐ Feed lightly, once a day, and siphon out anything uneaten
- ☐ Locate an aquatic vet now, while nothing is wrong, and write the number down

Week 2 to 3

- ☐ Move to the normal routine: 25 to 50 percent water change weekly, testing on the same day each week
- ☐ Settle the feeding schedule and portion, and start the weekly fasting day if you use one
- ☐ Start the Owner Log on page 35 with your weekly readings

Week 4

- ☐ Review the log. Readings should be stable and boring by now; a drifting nitrate means feeding or stocking needs adjusting
- ☐ Photograph the fish again and compare with day 1 for growth, color and fin condition
- ☐ Introduce the first enrichment items from page 37

Days 31 to 90

- ☐ If you quarantined a new fish separately, the 2 to 4 week minimum is up; 4 to 6 weeks is safer before it joins an established tank
- ☐ First filter maintenance: rinse mechanical media in removed tank water only, and leave the biological media alone
- ☐ Look back at the log for slow drift in nitrate, pH or temperature rather than single readings
- ☐ Fill in the pet-sitter sheet on page 34 before you need it, and check your test kit's expiry date

Symptom quick reference

Test the water before acting on any row in this table. It is the answer more often than anything else here.

SYMPTOM	LIKELY CAUSE	ACTION
Gasping at the surface, gills laboring	Low oxygen, or ammonia and nitrite burning the gills	Test now, 50% water change, add surface agitation (page 23)
Red or purple streaks in fins, bloodshot gills	Ammonia or nitrite poisoning	50% water change now, daily until 0 (page 23)
White spots like grains of salt	Ich	Treat the whole tank, 3 weeks minimum (page 23)
Fine gold or rust dust over the skin	Velvet	Treat the tank, dim the lights (page 25)
Flashing and rapid gilling, nothing visible	Flukes, or an early water quality problem	Test first, then praziquantel per vet guidance (page 25)
Frayed or receding fins	Fin rot, or nipping in a mismatched group	Water change, antibacterial if it advances (page 24)
Cotton-like white tufts	Fungus on an existing injury or stress point	Find the injury, treat antifungal (page 24)
Scales raised in a pinecone pattern	Dropsy, usually organ or kidney failure	Aquatic vet promptly. Poor prognosis (page 24)
Floating, sinking, swimming at an angle	Swim bladder disorder, usually diet-linked	Fast 24 to 48 h, then a de-shelled pea, switch to sinking food (page 25)
Long white stringy waste	Internal parasites, especially with weight loss	Vet or targeted antiparasitic, do not treat blind (page 26)
Long pale trailing waste	Constipation or overfeeding	Fast, then a pea, then soaked or sinking food (page 26)
Thread-like projection with red attachment point	Anchor worm	Vet guidance, do not pull it out (page 25)
One or both eyes protruding	Injury if one, systemic infection if both	Test water, then vet (page 25)
Open sores or red ulcers	Bacterial ulcers, water quality driven	Water change, then vet (page 25)
Sudden dark patches after a spike	Healing ammonia burns	Keep water perfect, they fade (page 23)
White pimples on gill plates and pectoral fins in spring	Breeding tubercles on a mature male, not disease	Nothing (page 20)
Chasing and nudging in spring	Spawning behavior	Normal. Separate only if a fish is exhausted or injured (page 26)
Cloudy white water in a new tank	Bacterial bloom during cycling	Normal. Do not add fish, do not deep clean (page 11)
Any change with no obvious cause	Several possible causes	Vet, and bring the Owner Log (page 35)

Daily, weekly & seasonal routine

Daily

- ☐ Feed the measured portion, once or twice, and siphon or net out anything uneaten
- ☐ Check the filter is running and the surface is still moving
- ☐ Quick visual check: fins open, breathing steady, color normal, no spots, all fish accounted for

Weekly

- ☐ Test ammonia, nitrite, nitrate and pH on the same day each week, and log it on page 35
- ☐ 25 to 50 percent water change, siphoning the substrate as you go
- ☐ Wipe the front glass and check the temperature
- ☐ Optional fasting day, particularly for fancy varieties

Monthly

- ☐ Rinse mechanical media in removed tank water. Never the tap, never all media at once
- ☐ Check the impeller and intake for buildup, and clean the airline and air stone
- ☐ Review the Owner Log for slow trends, not just single readings

Every 3 months

- ☐ Trim and thin plants, and check every piece of decor for new sharp edges or instability
- ☐ Photograph each fish from the side and above to compare growth and body shape over time

Seasonal

- ☐ Summer: watch for temperature climbing past 75°F (24°C). Warm water holds less oxygen, so increase surface agitation
- ☐ Pond only: scale feeding down with temperature, and stop below about 39°F (4°C)
- ☐ Spring: expect chasing and a few days off food in a mature mixed group

Yearly

- ☐ Replace the test kit; reagents lose accuracy after their expiry date
- ☐ Replace airline, air stones and worn hoses, and service or replace the filter impeller

Power outages, travel & transport

For a fish, a power cut is an oxygen problem first and a temperature problem a distant second.

When the pumps stop, the water stops moving and gas exchange stops with it. The fish will use the available oxygen faster than a still surface can replace it, and the filter's bacteria colony begins dying within hours without flow. A goldfish at room temperature is in far less danger from cold than a tropical fish would be, which is why aeration is essentially the whole job here.

Do

- Run a battery-powered air pump. A single deep-cycle unit can keep a tank aerated for a day or more, and models with a power-save cycle stretch that further.
- With no air pump, scoop out a cup of water and pour it back from a height every hour or two. Crude, but it genuinely moves oxygen into the water.
- Stop feeding. Uneaten food and waste consume oxygen and produce ammonia with no filter to process it.
- If the outage runs past a few hours, take the biological media out of the dead filter and put it in an aerated container or directly in the tank so the bacteria survive.

Don't

- Don't restart a filter that has been off for many hours without rinsing the media first. Stagnant media dumps a slug of decomposed waste straight into the tank.
- Don't do a large water change in the dark with untested, unconditioned water.
- Don't wrap or insulate the tank for a goldfish the way you would for a tropical species. Room temperature is well within range.
- Don't use candles, a gas stove, or any open flame near the tank.

Leaving town

TRIP LENGTH	WHAT IS ACTUALLY NEEDED
1 to 3 nights	Nothing. A healthy adult goldfish in an established tank is comfortable fasting for 3 to 4 days. Do a water change before you leave and skip the automatic feeder entirely.
4 to 7 nights	A visit or two to check the filter is running and nothing has died. Pre-portion the food into daily containers so a sitter cannot overfeed. Beyond about 5 days without food, condition starts to suffer.
Over a week	A sitter doing a real visit every two or three days, with one water change if the trip runs long. Walk them through the routine in person first.

Transport and vet trips

- Move the fish in a bag or sealed container of its own tank water, roughly one third water and two thirds air, inside an insulated cooler or box so it stays dark and the temperature stays stable.
- Do not feed for 24 hours before a move. Less waste in the water means less ammonia during the trip.
- For a journey over an hour, a battery air pump with an air stone in the container is worth the trouble, and a second bag as backup against a leak costs nothing.
- On arrival, acclimate as you would a new fish: float, mix, then net across, as on page 13.
- Never leave the container in a parked car, hot or cold, even briefly.

FILL IN AND LEAVE BY THE TANK

Pet-sitter sheet

Everything a sitter needs on one page. Walk them through it once in person before the first solo visit.

Tank volume and number of fish _____

Fish names or descriptions _____

My phone _____

Aquatic vet _____

Local fish store _____

Backup contact _____

Each visit

- | | |
|--|---|
| ☐ Check the filter is running and the water surface is moving | ☐ Net out any uneaten food after a couple of minutes |
| ☐ Count the fish and check each one is swimming normally | ☐ Check the temperature reads between 65 and 75°F (18 and 24°C) |
| ☐ Feed ONE pre-portioned container only. Do not add extra, whatever the fish do | ☐ Wash your hands afterward. Don't rinse anything in the kitchen sink. |

Please don't

- Feed more than the one pre-portioned container per visit. Goldfish beg constantly and always look hungry. Overfeeding is the single most likely way this goes wrong.
- Change, clean or unplug the filter, or top the tank up with untreated tap water.
- Tap the glass, move decor, or add anything at all to the water.

CALL ME IF

The filter is silent or the surface is still. A fish is gasping at the surface, lying on the bottom, floating oddly, or covered in spots. Any fish has died. The water has turned cloudy or smells bad. The power has been out for more than a few hours. If I don't answer, call the vet number above.

IF A FISH DIES WHILE I'M AWAY

Net it out and leave the rest of the tank alone. Do not do a big water change or clean the filter to compensate. Take a photo, message me, and check the remaining fish at the next visit.

Owner log

One row a week, filled in on test day. Photocopy this page, or track digitally. Consistency matters more than the format.

DATE	AMMONIA	NITRITE	NITRATE	PH	TEMP	% CHANGED	APPETITE, BEHAVIOR, NOTES

WHY BOTHER LOGGING

For a fish, the water readings are the vital signs, and unlike a weight they take two minutes and no handling. Almost every condition in this package announces itself in the numbers before it is visible on the fish. Sixteen rows is four months, which is long enough to see a real trend rather than a bad day, and it is the single most useful thing you can hand an aquatic vet.

Equipment & maintenance log

Equipment log

ITEM	INSTALLED	REPLACE OR CHECK BY	DONE
Tank, volume and date set up			
Filter, model and rated turnover			
Filter media last rinsed or replaced			
Air pump, airline and air stone			
Test kit, with expiry date			
Thermometer battery or replacement			

Twelve-month planner

Tick the box in the month you did it. Start with whichever month you're in.

TASK	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Rinse mechanical media	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Clean impeller and intake	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Trim plants, check decor	☐			☐			☐			☐		
Photograph each fish	☐			☐			☐			☐		
Replace test kit reagents	☐											☐

Incident & treatment log

Record every medication, salt dose and water emergency. Doses accumulate, and a vet will ask.

DATE	PROBLEM AND READINGS	WHAT WAS ADDED, AND HOW MUCH	OUTCOME

Enrichment checklist & log

Setup

- ☐ An open swim lane running the length of the tank
- ☐ Planted or structured cover at both ends, live or artificial
- ☐ Sand or fine smooth gravel deep enough to sift
- ☐ Sinking food as the default presentation

Monthly review

- ☐ Rearrange one piece of hardscape or planting
- ☐ Check nothing has developed a sharp edge or a fin trap
- ☐ Try one new safe food from pages 16 to 18
- ☐ Spend five minutes on a training session using the daily ration

Enrichment log

DATE	ACTIVITY	HOW IT WENT, AND WHETHER TO REPEAT

ENRICHMENT IS NOT EXTRA FOOD

Every foraging and training idea in this package uses the fish's existing daily ration, divided differently. A goldfish will happily eat itself into a buoyancy problem, so scattering pellets or running a training session means feeding the same amount in a more interesting way, not adding a second meal on top.

Glossary

Ammonia	The first waste product of fish waste and decaying food, and highly toxic. The target reading is 0 ppm in a cycled tank.	Ich	White spot disease, caused by <i>Ichthyophthirius multifiliis</i> . Only its free-swimming stage can be treated, which is why treatment must outlast the visible spots.
Bioload	The total waste output a tank's filtration and water volume have to process. Goldfish have an unusually high bioload for their size.	New tank syndrome	Ammonia and nitrite poisoning in a tank stocked before it finished cycling. The most common cause of early goldfish death.
Chloramine	A chlorine and ammonia compound used in many municipal water supplies. Unlike chlorine it does not gas off if water is left standing, so a conditioner is required.	Nitrate	The end product of the nitrogen cycle. Far less toxic than ammonia or nitrite, but keep it under about 20 ppm with water changes.
Cycling	Establishing the bacteria colony that converts ammonia to nitrite and nitrite to nitrate. Takes roughly four to six weeks in a new tank.	Nitrite	The intermediate product of the nitrogen cycle, also highly toxic. The target reading is 0 ppm.
Dechlorinator	A water conditioner that neutralizes chlorine and chloramine in tap water before it reaches the fish.	Quarantine	Housing a new or sick fish in a separate cycled tank, 2 to 4 weeks minimum and 4 to 6 weeks preferably, before it meets an established group.
Dropsy	Not a disease but a sign: fluid swelling that lifts the scales into a pinecone pattern, usually from organ or kidney failure.	Single-tail	The streamlined varieties: common, comet and shubunkin. Larger, faster and much better suited to a pond than a tank.
Fancy goldfish	The round-bodied ornamental varieties (oranda, ranchu, fantail, ryukin, moor). Slower swimmers, most prone to buoyancy problems.	Slime coat	The living protective layer over a goldfish's scales. It takes 24 to 48 hours to regenerate if damaged by handling.
Flashing	Rubbing or scraping the body against surfaces. A sign of irritation from parasites or poor water quality.	Stunting	Suppressed body growth from cramped space and chronic ammonia exposure while organs and skeleton keep developing. A welfare problem, not a size feature.
GH and KH	General hardness (dissolved minerals) and carbonate hardness (the water's ability to resist pH swings). Low KH lets pH crash suddenly.	Swim bladder	The internal organ controlling buoyancy. Problems with it cause floating, sinking, or swimming at an angle.
Gravid	Carrying eggs. A gravid female often looks fuller and asymmetrical viewed from above.	Tubercles	Rough white breeding spots on the gill plates and pectoral fin rays of a mature male in breeding condition. Often mistaken for ich.
Hospital tank	A small, separately cycled tank used for treating a sick fish, so medications do not reach the display tank's bacteria colony.	Turnover	How many times per hour a filter processes the tank's total volume. Aim for 4 times minimum, 5 to 10 preferred.

REFERENCE

Sources, version history & about

Where the numbers in this package come from, so you can check them yourself.

Veterinary and scientific sources

- Merck Veterinary Manual: Parasitic Diseases of Fish, for the ich life cycle and its treatable stage, and for the praziquantel, diflubenzuron and salt figures on page 25.
- PetMD: Goldfish Care Sheet, for temperature, pH, adult size, lifespan and the signs-of-illness list.
- Sullivan et al., on goldfish habitat preference, summarized by Companion Animal Psychology: the roughly 90 percent planted versus 10 percent barren finding on page 19.
- Smith SA and Gray J, substrate particle size and goldfish foraging behavior, Universities Federation for Animal Welfare.
- Guinness World Records: oldest goldfish, for the 43-year figure on page 4.

Husbandry references

- The Goldfish Tank, on disease identification, slime coat damage from handling, and water parameters.
- Toledo Goldfish: identifying goldfish diseases.
- Aquarium Co-Op and Florida Sea Grant, on keeping an aquarium safe during a power outage, for page 33.
- Aquarium Science, on fish-in cycling and the 0.25 ppm water change threshold on page 9.
- University of Florida IFAS Extension: freshwater ornamental fish commonly cultured in Florida.
- Beastly Facts goldfish guides: tank setup, feeding, health issues, handling, cost, enrichment, and the tank size and bowl myth deep dive, current at the time of publication.

Version history

VERSION	TEMPLATE	WHAT CHANGED
2.0 · Sep 2026	t3	Rebuilt on the t3 template and expanded from 21 to 39 pages. Filtration, cycling, water chemistry and water changes each got their own page, with a full parameter table including pH, GH and KH, and a fish-in cycling plan for tanks already holding a fish. Added a tank diagram, separate staple, protein and never-feed charts, varieties and sexing, a growth reference and body condition guide, quarantine and hospital tank guidance, expanded health coverage to flukes, anchor worm, velvet, popeye and ulcers with veterinary treatment figures, a reading-waste table, a power outage plan, a travel and transport page, a pet-sitter sheet, an incident and treatment log, and this sources page. The owner log now records water readings rather than weight, which is the practical vital sign for a fish. Metric units throughout.
1.0 · Aug 2026	pre-t3	First edition, 21 pages.

This package is a husbandry reference compiled from the sources above and cross-checked against current veterinary guidance at the time of publication. It is not a substitute for an aquatic veterinarian, and nothing in it should delay a vet visit. Medication doses quoted here are from published veterinary references and are included so you can discuss them with a vet, not so you can dose a tank unsupervised. Free companion guides, updated as guidance changes, are at beastlyfacts.com.

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